Sifatul Anindho

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1. Introduction

Graduate researcher interested in applied ML (multimodal, computer vision, and NLP) to support context-aware human–AI interaction. Open to internships that align with my research interests.

2. Education

- **Colorado State University** — *Expected May 2029*
  Doctor of Philosophy: Computer Science — Fort Collins, Colorado
  - GPA: 4.00/4.00
- **Colorado State University** — *Expected Jul 2026*
  Master of Science: Computer Science — Fort Collins, Colorado
  - GPA: 4.00/4.00
- **Michigan State University** — *May 2024*
  Bachelor of Science: Computer Science — East Lansing, Michigan
  - GPA: 3.81/4.00

3. Research Positions

- **IMPACT Lab, Colorado State University** — *August 2024 - Present*
  Graduate Research Assistant, Advisor: Dr. Nathaniel Blanchard — Fort Collins, Colorado
  - Cognitive and affective state modeling in multimodal, multiparty settings
- **HLR Lab, Michigan State University** — *May 2024 - July 2024*
  Undergraduate Research Assistant, Advisor: Dr. Parisa Kordjamshidi — East Lansing, Michigan
  - Natural language tool for crafting complex neuro-symbolic models
- **Liu Lab, Michigan State University** — *May 2023 - August 2023*
  Undergraduate Research Assistant, Advisor: Dr. Kevin Liu — East Lansing, Michigan
  - Empirical performance of algorithms for phylogenetic estimation

4. Research Projects

- **FACT: Friction for Accountability in Conversational Transactions** — *August 2024 - August 2025*
  Funding: DARPA
  - Designed retrospective cued-recall labeling framework to synchronize self-reported cognitive-affective trajectories with facial action units
  - Developed alignment procedures between self-reported internal states and AI-assistant actions to study intervention effectiveness.
  - Developed temporal alignment algorithms for uncertain state boundaries and implemented pipelines for fuzzy sequence labeling.
  - Explored sequence modeling approaches that account for the temporal uncertainty in cognitive-affective dataset
  - Built multimodal feature extraction systems for real-time inference, including OpenFace3-based facial behavior streams
- **PARADIGM: Platform Accelerating Rural Access to Distributed and Integrated Medical Care** — *May 2025 - Present*
  Funding: ARPA-H
  - Developed multimodal feature extraction pipelines for facial action units, enabling state estimation models for cognitive load in clinical tasks.
  - Developed object-tracking pipelines for ultrasound-probe localization and probe-holder hand-tracking using RealSense SDK and MediaPipe.
  - Analyzed practitioner's facial expression data to benchmark open-source facial expression recognition tools for cognitive-affective states during DVT scans

5. Peer-Reviewed Publications

C=Conference, J=Journal, W=Workshop, D=Doctoral Consortium

[W.1] Sifatul Anindho, et al. (2025). **A Methodological Framework for Capturing Cognitive-Affective States in Collaborative Learning**. In *Interactive Workshop: Multimodal, Multiparty Learning Analytics (MMLA) at the conference Educational Data Mining (EDM) 2025*

[C.1] Sifatul Anindho, et al. (2025). **An Exploration of Internal States in Collaborative Problem Solving**. In *International Conference on Human-Computer Interaction 2025*, (pp. 135-150). Cham: Springer Nature Switzerland.

In Press (Accepted):

[W.2] Sifatul Anindho, et al. (2026). **Evaluating Web-trained Facial Expression Recognition in Collaborative Problem-Solving**. In *IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW): Computer Vision for Education 2026*

[W.3] Ethan Seefried, et al. (2026). **CSU101: An Educational Dataset for Introductory Computer Vision**. In *IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW): Computer Vision for Education 2026*

[C.2] Sifatul Anindho, et al. (2026). **Ordered Network Anlaysis of Epistemic Emotions during Collaborative Problem Solving**. In *International Conference on Artificial Intelligence in Education (AIED) 2026*

[C.3] Videep Venkatesha, et al. (2026). **Using LLMs to Annotate Pedagogical Moves: You Know What I Mean?**. In *International Conference on Artificial Intelligence in Education (AIED) 2026*

[C.4] Videep Venkatesha, et al. (2026). **Detecting Self-Reported Cognitive-Affective States in Collaborative Problem Solving from Prosodic and Linguistic Features**. In *International Educational Data Mining Society (EDM) 2026 Poster/Demo Track*

[D.1] Sifatul Anindho and Nathaniel Blanchard (2026). **Identifying and Modeling Epistemic Emotions in Collaborative Problem-Solving**. In *International Educational Data Mining Society (EDM) 2026 Doctorial Consortium Track*

6. Research Presentations

T=Talks, P=Posters, D=Demos

P.1 **Deep Vein Detection, Tracking, and Compression Guidance**, *ARPA-H Site Visit on Vectores of Intelligent Guidance in Long-Reach Rural Healthcare*, 2025.

[P.2] **Retrospective Cued Recall for Labeling Cognitive-Affective States in Collaboration,** *Graduate Student Showcase, Colorado State University*, 2025. [Accepted but unable to present due to conflict with D.1]

[D.1] **AI Guidance in Lower-Limb Ultrasound for Deep Vein Thrombosis**, *ARPA-H PI Meeting on Platform Accelerating Rural Access to Distributed and Integrated Medical Care*, 2025.

[D.2] **TRACE: Transparency, Reflection, and Accountability in Collaborative Exchanges**, *DARPA PI Meeting on Friction and Accountability in Conversational Transactions*, 2025.

[T.1] **An Exploration of Internal States in Collaborative Problem Solving,** *International Conference on Human-Computer Interaction*, 2025.

[P.2] **DNA Sequence Generation and Classification with Symbolic Domain Knowledge using DomiKnowS**, *Mid-Michigan Symposium for Undergraduate Research Experiences*, 2024

7. Research Software

- **Multi-Label Video Annotation Tool** — *October 2025*
  Projects: PARADIGM
  - A tkinter based local application to support fast annotation of videos for multi-label classification
- **Retrospective Cued-Recall based Annotation Tool** — *February 2025*
  Projects: FACT, PARADIGM
  - A web-hosted solution for retrospective-cued recall based annotation using self and probe caught

8. Teaching

- **Colorado State University** — *August 2024 - Present*
  Graduate Teaching Assistant — Fort Collins, Colorado
  - CS 445 - Introduction to Machine Learning - Spring 2026
  - CS 345 - Machine Learning Foundations and Practice - Fall 2025
  - CS 214 - Software Development, Spring 2025
  - CS 164 - Computational Thinking with Java, Fall 2024, Summer 2025 (as course design consultant)
- **Michigan State University** — *January 2022 - May 2024*
  Undergraduate Teaching Assistant — East Lansing, Michigan
  - CSE 102 - Algorithmic Thinking and Programming, Spring 22/23/24, Fall 22/23

9. Academic Service

- **Judge, CURC 2026** *April 2026*
- **Reviewer, CVPRW (CV4Edu) 2026** *March 2026*
- **Reviewer, ACL 2026** *February 2026*
- **Reviewer, ACL 2025** *March 2025*
- **Graduate Program Committee Representative, Colorado State University** *January - May 2026*

10. Selected Software Engineering position

- **Whirlpool** *January 2024 - May 2024*
  Senior Design Intern Benton Harbor, Michigan
  - Software development for a human machine interface for Whirlpool smart ovens

11. Selected Software Project

- **Lab Operations Scheduler for Graduate Student Assistants** *Sep 2025 - Dec 2025*
  Client: Colorado State University, open-sourced
  - Built an integer programming optimized system to assign lab operators to timeslots based on availability and preferences from 50 operators
  - Modeled multi-constraint scheduling inspired by NSF panel-assignment frameworks

12. Advanced Technical Skills

- **Programming Languages:** Python, R, JavaScript, Java, C/C++, MATLAB, SQL, Bash
- **Machine Learning & Data Science:** PyTorch, PyTorch Lightning, scikit-learn, TensorFlow/Keras, HuggingFace, pandas, NumPy, OpenCV, W&B, MLflow, Pydantic, Hyperopt, Jupyter, Colab
- **Multimodal & Experimental Systems:** OpenFace3, MediaPipe, Google ML Kit, ROS2, Unity, RealSense SDK, gradio
- **Web & Application Development:** Flask, FastAPI, Node.js, React.js, Socket.io, HTML/CSS, MongoDB, PostgreSQL, MySQL
- **Cloud & Distributed Systems:** AWS, GCP, Azure, SageMaker, Spark, Hadoop, Docker, Kubernetes, Git
- **Mathematical & Statistical Tools:** Probabilistic Modeling, Statistical Analysis, Time-Series & Sequential Analysis, Dimensionality Reduction, SAS, Data Visualization (matplotlib, seaborn, plotly)
- **Specialized Areas:** Multimodal Learning, Human-Centered AI, Cognitive-Affective Modeling, Human-Subject Experiment Design, Interactive System Prototyping

13. Selected Academic Distinctions

- **2nd Place Award in CSU Club Computer Vision Competition** *November 2024*
  Colorado State University
  - Awarded for performing second-best in a university-wide computer vision competition
- **Global Spartan Scholarship** *January 2021*
  Michigan State University
  - Awarded to top 10% achieving undergraduates in academics
- **Outstanding Cambridge Learner Award** *June 2018*
  University of Cambridge
  - Achieving the best score in the world in the AS Level Mathematics examination

References

1. **Dr. Nathaniel Blanchard, Colorado State University**
2. **Dr. Nikhil Krishnaswamy, Colorado State University**
3. **Dr. Anne M. Cleary, Colorado State University**